

Volume V · Chapter 25 — Learning for Control (Imitation/RL, sim-to-real) · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-V-Chapter-25-context.md for the AI interaction log.

Prerequisites: Ch 23, Ch 24.

Learning outcomes — the student can: explain imitation learning & RL for control; train a policy in simulation; reason about sim-to-real transfer and its risks; apply safety constraints/shields.

Labs (hands-on) — 14 h:

#	Lab	h
25a	Behavioral cloning on a sim manipulation task	4
25b	Train an RL policy in sim (reach/grasp)	5
25c	Domain-randomization experiment	3
25d	Add a safety constraint / shield to the policy	2

Datasets/tools: RL libraries (Stable-Baselines3 / Isaac Gym); the simulator.

Assessment: trained policy (**60%**); sim-to-real analysis (**20%**); quiz (**20%**). **Key**

decisions: IL vs. RL; reward design; safety shielding; when sim is insufficient.

References: plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **10** + Lab **14** = **24**.